TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Machine Operation: PLC and SCADA

The two software layers that run a modern textile machine: the PLC executing its ladder program a thousand times a minute at the machine, and SCADA above it — supervising, recording and letting one operator see a whole dyehouse at a glance.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 2 · LEVEL 3 TERM 2

1. Aim

To read and draw a ladder-logic program for a motor start/stop (seal-in) circuit, explain the PLC scan cycle and fail-safe wiring, and describe how a SCADA system supervises PLC-controlled textile machines — using a dyeing machine as the running example.

2. Concepts you need

- **PLC (Programmable Logic Controller):** a rugged industrial computer that reads inputs (pushbuttons, sensors, limit switches), executes a stored program, and drives outputs (motor contactors, valves, lamps). It replaced panels of hard-wired relays; changing the machine's behaviour means editing the program, not rewiring.
- **Ladder logic:** the program is drawn as rungs between two power rails, read left to right, top to bottom — contacts (inputs and internal states) on the left, coils (outputs) on the right. A rung's coil energises when a continuous path of "true" contacts connects the rails.
- **The scan cycle:** read all inputs → execute the program once → write all outputs → repeat. Typical cycle: a few milliseconds. Everything the PLC "sees" is a snapshot renewed each scan.
- **Seal-in (latching):** a momentary START button cannot hold a motor on by itself. The motor's own auxiliary contact, wired in parallel with START, keeps the rung true after the button is released — the contactor *seals itself in* until STOP breaks the rung.
- **Fail-safe wiring:** STOP and overload contacts are wired **normally closed (NC)** in the field, so the PLC input is normally TRUE and the rung breaks when the button is pressed — *or when the wire breaks*. A broken stop circuit stops the machine; a broken wire on a normally-open stop would silently remove the stop function. This is why the ladder rung shows the stop contact as a "pass" (examine-if-closed) instruction fed by an NC field device.

- **SCADA (Supervisory Control and Data Acquisition):** the software layer above the PLCs. It collects data from every connected machine, draws live mimic screens, raises and logs alarms, records trends in a historian database, and lets supervisors download recipes/setpoints (e.g. a dyeing program) to the machines. The PLC still runs the machine if SCADA fails — supervision is lost, control is not.

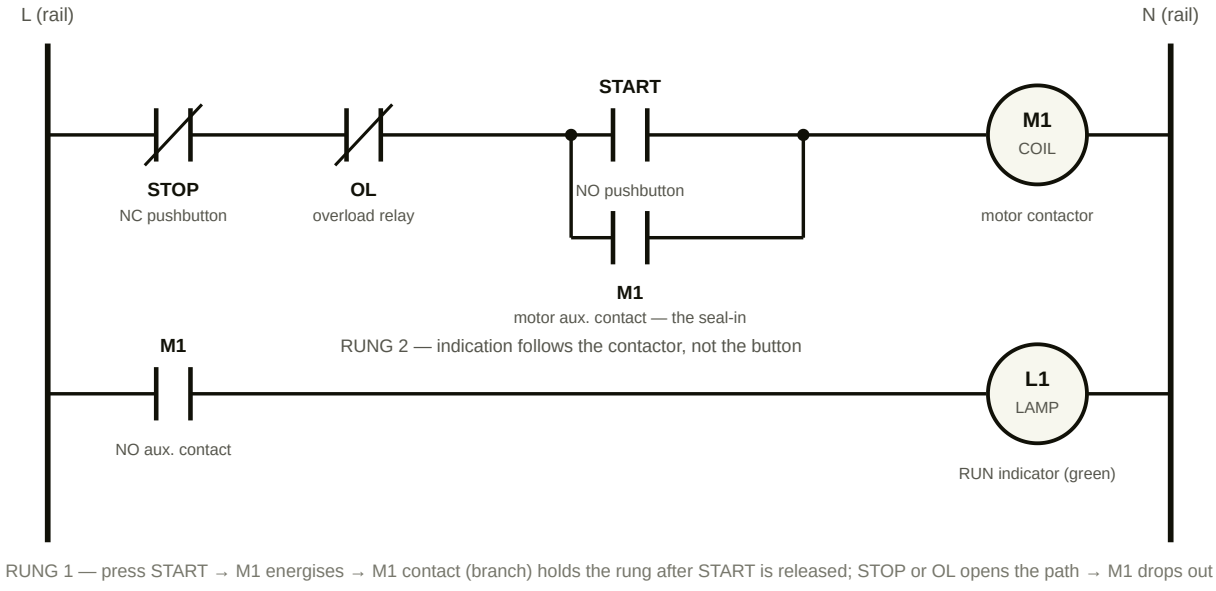


Figure 1 — Ladder program for a motor start/stop with seal-in. NC field devices (STOP, OL) appear as pass contacts: their input bits are TRUE in the healthy, un-pressed state, so a broken wire reads exactly like a pressed STOP.

3. Lab procedure

1. Enter the ladder program of Figure 1 in the lab's PLC software (or a ladder simulator) and download it to the training PLC.
2. Press START briefly: the motor output energises and *stays* energised after release. Identify, on the rung, exactly which contact is now carrying the rung.
3. Press STOP: output drops. Press START and simultaneously hold STOP: the motor must never start — explain why from the rung's series structure.
4. Simulate a broken STOP wire (disconnect the input): the motor cannot be started. Record this as the fail-safe property in action.
5. Trigger the overload input while running: the motor stops and will not restart until the overload is reset. Note how this differs from a STOP press.
6. Open the connected SCADA/HMI screen: start and stop the motor from the screen, force an alarm (high temperature on the demo dyeing vessel), and find the event in the alarm log and the trend display.

4. Worked example — reading the seal-in rung, scan by scan

Healthy state: STOP un-pressed and overload healthy, so inputs *STOP* and *OL* are TRUE; *START* and *M1* contacts are FALSE; coil M1 is off.

MOMENT	WHAT THE SCAN SEES AND DOES
Operator presses START	Next scan: STOP ✓, OL ✓, START ✓ (main branch) → rung true → **coil M1 energises**; contactor closes, motor runs, and M1's auxiliary contacts change state in the input image.
Operator releases START	START returns FALSE, but the M1 auxiliary contact (parallel branch) is now TRUE. Rung: STOP ✓, OL ✓, (START ✗ OR M1 ✓) ✓ → coil stays energised. **This is the seal-in.**
Operator presses STOP	STOP input goes FALSE → rung broken at the first contact → coil de-energises → M1 aux contact opens → the seal is broken. Releasing STOP restores the input, but both parallel branches are now FALSE, so the motor stays off until START is pressed again.
Supply fails and returns	Coil drops during the failure; on restoration both branches are FALSE — the machine does **not** restart by itself. Deliberate: an unexpected self-restart is a safety hazard, so re-starting must be a human act.

Scan-time check: the lab PLC reads inputs in 2 ms, executes this program in 5 ms and writes outputs in 1 ms — a scan of 2 + 5 + 1 = **8 ms**. A signal pulse shorter than one scan can fall entirely between two input reads and be missed; that is why real stop/start signals are held states or latched in hardware, never millisecond pulses.

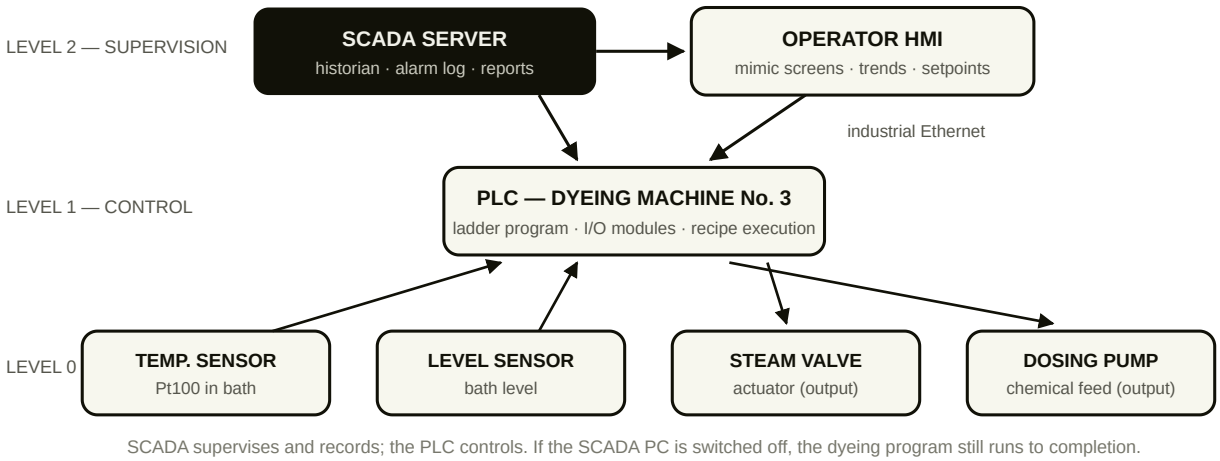


Figure 2 — The three levels of machine control for a PLC/SCADA dyeing machine. Sensor and actuator signals are hard-wired to the PLC; only data travels upward.

5. Practice tasks

1. A chemical dosing pump must run while a pushbutton station's START is pressed, stop on STOP, and stop automatically and stay stopped when the tank's high-level float switch opens. Describe the rung (contacts in order, coil) and state which contacts are wired NC in the field and why.
2. An operator complains that a machine "sometimes ignores" a very quick jab at a sensor that produces a 3 ms pulse. The PLC scan time is 8 ms. Explain the complaint and name the standard cure.
3. List four things the SCADA historian gives a dyehouse manager that the PLCs alone cannot, and one risk of relying on SCADA that the level structure of Figure 2 is designed to avoid.

Answer key

1. Rung: rail → **STOP** (pass contact; field device wired NC) → **HIGH-LEVEL float** (pass contact; field switch NC, opens at high level) → parallel branch of **START** (NO) and **P1 aux** (seal-in) → coil **P1** (pump). STOP and the float switch are wired NC so that a broken wire or a tripped switch halts the pump — the dangerous failure (overfilling) and the wiring failure produce the same safe state. Because the float sits in the series path *before* the seal-in branch, opening it breaks the seal: the pump stays off even after the level falls, until START is pressed again.
2. The PLC samples its inputs once per scan. A 3 ms pulse that begins and ends between two input reads (8 ms apart) never appears in the input image, so the program never sees it — the machine is not faulty, the signal is shorter than the sampling interval. Cure: latch the event in hardware (a latching relay or a sensor with pulse-stretching output) or use a high-speed counter/interrupt input designed to catch short pulses.
3. From the historian: (i) the temperature–time curve of any past batch, for reproducing a shade or investigating a complaint; (ii) alarm history with timestamps, showing which machine faults recur; (iii) utility consumption per batch (steam, water) for costing; (iv) proof of process for buyer audits — the recorded curve shows the dyeing actually followed the program. The avoided risk: if control lived in the SCADA PC, a PC crash or network failure would stop production; with control in the PLCs (Level 1) and SCADA only supervising (Level 2), a supervision failure degrades visibility, not production.
